Question Paper

Continuous Assessment Examination - II (CAE-II)

Programme: B. Tech | **Year: First** | **Semester: I**

Max Marks: 20 | **Duration: 60 mins.**

Course Name: Engineering Mathematics | **Course Code: MT101**

Instructions:

1. Write the solution in detail.
2. Clearly show all steps of your work, including necessary derivations and calculations.
3. Read each question carefully before attempting to answer.
4. Students must label their answers with the corresponding question number, such as Q. No. 1A, Q. No. 2C, etc.

Note: This question paper addresses below Course Outcomes (CO's):

CO3	Utilize Matrix Algebra for Computations and Analysis
CO4	Solve Linear Equation Systems by using matrix method

Q. No.	Sub Q. No.	Question	Marks
1		Solve any **TWO**	
	A	find the inverse of the following matrix (if it exists): $A = \begin{pmatrix} 2 & 1 & 0 \\ 3 & 4 & 1 \\ 1 & 2 & 3 \end{pmatrix}$	05
	B	a) Find The Eigen Values of matrix of order 2 × 2 such that trace of matrix - 2, and determinant of matrix is -35 b) Define 1) idempotent matrix, 2) involuntary matrix	05
	C	Verify Cayley-Hamilton Theorem and find A^{-1} of matrix $A = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 1 & 2 \\ 1 & 2 & 1 \end{bmatrix}$	05
	D	Compute the product of the following matrices: $A = \begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 4 \\ 2 & -3 \end{pmatrix}$ Find a) Find A + B b) Find A - B c) Find A × B d) Determinant of A, B e) Determinant AB	05
2		Solve any **TWO**	
	A	Use Cramer's Rule to solve the following system of linear equations: x + y + z = 6 2x - y + 3z = 14 x + 2y - z = 2	05
	B	Define rank of matrix and find the rank of the matrix: $A = \begin{pmatrix} 2 & 1 & 0 \\ 2 & 4 & 6 \\ 1 & 0 & 1 \end{pmatrix}$	05
	C	Solve the following system of equations using the matrix inversion method: 2x + y + z = 5 x - 3y + z = 0 x + y - 2z = 2	05
	D	Find the row echelon form also find rank and nullity of given matrix. Also Define Nullity of matrix. $A = \begin{bmatrix} 3 & -1 \\ -1 & 5 \end{bmatrix}$	05